

I am a software engineer with 7+ years of experience building scalable backend and data systems across geospatial, energy, and fintech domains. I am highly proficient in Python, Postgres and asynchronous programming and have built complex applications and end-to-end data processing pipelines with successful business outcomes. I have scaled a digital bank's loan portfolio, built an end-to-end hydropower data processing pipeline for a renewable energy company and asynchronous satellite image data processing workflows for farmers around the world.

Senior AWS Engineer (Contract)	ProductFlight E-Commerce, European Union	2026
- Integrated OIDC with Pulumi, AWS Cognito and Microsoft Azure Entra ID for Single-Sign-On to expand potential user base - Improved GitHub Actions continuous-integration pipeline by optimizing cache usage using Python uv and workflows - Integrated product information management systems (PIM) with LLMs using LangChain, MCP, FastAPI to reduce analyst toil - Expanded and improved application observability using Betterstack, Lambda, FastAPI		
Fullstack Geospatial Developer (Remote)	DataFarming Precision Agriculture, Australia	2022–Present
- Led the commercial deployment of crop frost damage analytics for continent-scale satellite imagery - Architected and implemented the optimization of legacy web map tiling endpoints using cloud-optimized GeoTIFFs that reduced user latency for high-resolution imagery by 70%, unblocking the scaleup of the product - Led the implementation of data pipelines and asynchronous event-driven microservices and RESTful API for large-scale soil health data as a web-map product using <i>FastAPI</i>, <i>PostGIS</i>, <i>SQS</i>, <i>S3</i>, <i>Python</i>, <i>ECS</i> and <i>ElastiCache</i> - Scaled the asynchronous processing and import of point data for vector tiles by optimizing the database loading for 20 million data points from >30 minutes during prototyping to <5 minutes using the <i>Python Dask</i> framework and improved UX - Implemented the core services for on-demand processing of geospatial file format conversion using <i>FastAPI</i> for the endpoint and <i>Lambda</i> for compute and optimized latency for complex data processing to the target of 3–5 seconds for 95% of requests - Prototyped the web-map components for high-volume point data using vector tiles of the <i>Mapbox GL JavaScript</i> framework for the frontend team - Implemented the event-driven architecture and optimized the application's geostatistics modelling service using <i>FastAPI</i>, <i>S3</i>, <i>SQS</i> and <i>ECS</i>. Automated complex data processing, model training and deployment. Improved execution from 4 hours to less than 10 minutes in production - Implemented and deployed the data warehouse and Extract-Load-Transform service and RESTful API using <i>Postgres</i>, <i>FastAPI</i>, <i>ECS</i> to enable visibility of product business performance - Improved <i>ElastiCache</i> caching reliability by refactoring key locking mechanism to improve data consistency and reliability - Introduced retry mechanisms for data imports from vendor APIs to improve application resiliency - Expanded real-time application monitoring and logging using <i>Sentry.io</i> to identify and address bottlenecks in microservices - Practiced <i>Test-Driven Development (TDD)</i> using Python's <i>Pytest</i> framework and <i>LocalStack</i> to implement unit and integration tests and improve stability and reliability of the application backend when introducing changes - Optimized infrastructure and deployment using <i>Make</i>, <i>Docker</i>, <i>CloudFormation</i> for infrastructure-as-code, <i>CodePipeline</i> and <i>CodeBuild</i> for CI/CD to improve developer experience - Developed the disaster recovery process for catastrophic data loss for the AWS RDS <i>Postgres</i> cluster for operational resiliency		
Backend Developer Consultant (Contract)	El Observatorio de Manila Non-Profit, Philippines	2022
- Led the implementation of image processing pipelines of satellite images using geospatial modules in <i>Python</i> and <i>Docker</i> for post-catastrophe rapid damage assessment - Implemented RESTful APIs that served extracted flood data from rasters using <i>Flask</i> and <i>Docker</i> for frontend clients - Implemented a command-line interface in <i>Bash</i> to simplify execution for analysts - Upskilled and mentored junior analysts in version control, software testing and DevOps		

Data Management Lead	SN-Aboitiz Power Energy, Philippines	2021–2022
- Integrated operations and information technology (OT-IT) systems of hydroelectric power plants to the cloud using <i>Python</i>, <i>SMB protocol</i>, <i>Google Object Storage</i>, <i>BigQuery</i> and <i>Google Sheets</i> and reduced analyst toil - Planned the data engineering and management infrastructure roadmap and secured the implementation budget for 2022 - Enhanced and expanded the data warehouse schema by implementing data cataloguing practices across the company - Upskilled teams with SQL to improve data warehouse utilization by 8% in the next quarter and improved data provenance - Prototyped the data integration of the Oracle ERP system with BigQuery using Cloud Composer and collaborated with the finance team for the data governance of the company's chart of account - Mitigated security risks to the data warehouse by implementing the principle of least privilege using <i>Google Cloud IAM</i>		
Big Data Engineer	CIMB Bank PH Fintech, Philippines	2019–2021
- Led and mentored a team of 3 big data engineers by conducting architecture and code reviews and ticket management - Set up and implemented the security hardening of the <i>MongoDB</i> cluster to automate know-your-customer operations. Implemented indexing to optimize lookups of unstructured customer data for the loan underwriting team - Automated workflows in banking operations by implementing ETL using <i>CentOS</i>, <i>MySQL</i>, <i>Hive/Hadoop</i>, <i>Bash</i> and <i>Python</i> - Reduced ETL latency from 12 hours to <30 minutes by using <i>MySQL</i> composite indexes and daily delta loading - Prototyped market campaign data pipeline using <i>Airflow</i>, <i>Postgres</i> and <i>HDFS</i> for personalized messaging to customers - Led the implementation of on-premise <i>GitLab</i> and <i>JupyterHub</i> to improve the developer experience of internal teams - Led the data integration of customer transaction data with credit bureau partners by building complex transformations in <i>Hive</i> - Participated in cybersecurity incident response by extracting and analyzing transaction data and logs		
Analytics Specialist	Accenture Business Outsourcing, Philippines	2017–2019
- Developed a task scheduling tool using <i>Python</i> to optimize agent scheduling for a global legal services client that improved team utilization by 50% after 1 month of deployment - Automated fullstack data processing using <i>Python</i> and built a dashboard in <i>Power BI</i> to eliminate manual reporting and improve operational awareness of leads - Performed exploratory data analysis and uncovered \$57 million Australian dollars in accelerated working capital impact (KPI) for a large energy retailer		
GIS and Remote Sensing Technician (Contract)	El Observatorio de Manila Non-Profit, Philippines	2014–2017
Automated complex geospatial data processing and machine learning workflows of senior scientists using <i>Python</i>, <i>QGIS</i>, <i>ArcGIS Pro</i>, <i>ArcPy</i>, <i>ENVI</i> and <i>Jupyter Notebooks</i> to improve analysis turnaround times for disaster response		

Projects

- Routes.ph:** Fullstack personal project for bike commuting using OpenStreetMap data. Architected and implemented infrastructure using *Terraform* and *Ansible*. Implemented frontend web-map using *Vue* and *Leaflet* and the backend routing engine using *GraphHopper*, API using *Flask* and database using *Postgres* and *PostGIS*
- SpaceCrop Technologies:** Hungary-based agritech startup. Part-time backend developer and DevOps consultant to automate the build-up and teardown of backend infrastructure in *DigitalOcean* using *Terraform* and *Ansible*

Skills

- Languages:** Python, SQL, Bash, C, JavaScript
- Frameworks:** FastAPI, Flask, Vue, Mapbox GL
- Cloud Services:** Google Cloud, AWS, DigitalOcean
- Storage:** Postgres, MySQL, HDFS, MongoDB, BigQuery, S3, Object Storage
- DevOps:** Docker, Git, CI/CD, Linux, FTP/SFTP, SSH, Terraform, Ansible, OpenAPI, Swagger, CloudFormation, Sentry

Education

- Master of Science in Geography (thesis pending)**, University of the Philippines, Diliman
- Basic and Advanced GIS**, NAMRIA Geomatics Training Center, DENR, Taguig City
- Bachelor of Arts in Political Science, Minor in Philosophy**, Ateneo de Manila University